



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Roll no.....

Name of the Course and semester: BCA/BCA(AI/DS)/Bsc(IT)

Name of the Paper : Data Communication and Computer Networks

Paper Code: TBC 404/TBD 403/TBI 404

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions
- (ii) Each question carries 10 marks.
- ~~(iii) Please specify COs against each question.~~

Q1. (10 Marks)

- a. Compare Hub and Switch as networking devices. Analyze their working principles and explain why switches provide better performance than hubs in modern networks. CO1

OR

- b. Define Amplitude Shift Keying (ASK). Explain its principle of operation with a neat labeled diagram. Critically analyze its advantages and limitations, and discuss its suitability for Digital to Analog encoding. CO2

Q2. (10 Marks)

- a. Identify and explain the three major types of wired transmission media used in data communication networks. Analyze their internal structure and its application. Support your answer with neat labeled diagrams. CO2

OR

- b. Explain the need for multiplexing in Data Communication and Networking (DCN). Discuss how multiplexing improves cost efficiency. Describe Frequency Division Multiplexing (FDM) in detail. CO1

Q3. (10 Marks)

- a. Write short notes on following : CO1

- (i) Gateway
- (ii) Simulation
- (iii) Switches
- (iv) NIC
- (v) IP addressing

OR

- b. Define a Computer Networking Model. Describe the different types of networking models with their neat labeled diagrams.

Q4. (10 Marks)

- a. Compare Intranet, Internet, and Extranet as networking environments. Analyze at least ten distinguishing parameters. CO1

OR

- b. Explain Polar Signaling as a digital line coding technique. Analyze the functional role of Polar NRZ and Polar RZ. Support your answer with suitable diagrams for this representation. CO2



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Q5.

(10 Marks)

- a. Differentiate Unicasting, Broadcasting, and Multicasting as data delivery mechanisms. CO3

OR

- b. Define Pulse Code Modulation (PCM). Explain each stage of PCM in detail and analyze how quantization error affects the accuracy of the digital signal. CO2